Algebraic Geometry

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0.1 Plan

1. Basics of sheaves of topological spaces
2. Definitions of schemes and of morphisms
3. Properties of schemes (the algebraic geometry analogues of compactness)
4. Rapid introduction to the cohomology of sheaves

0.2 Resources

1. Course webpage - Google Dhruv
2. Books:
   - Hartshorne - Algebraic Geometry, Chapter 2 and $\varepsilon\times$Chapter 3
   - Vakil - Rising Sea, good online (free) alternative
   - Eisenbud-Harris - Geometry of schemes
3. Four examples sheets on webpage
4. SAGES - Look on Dhruv's webpage

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0.3 Why scheme theory?

Moduli theory – It is better to study families of varieties than one at a time. Or better, all varieties of a given type at the same time.

Examples of moduli: The set of all lines in $\mathbb{P}^2$. A line in $\mathbb{P}^2$ is

$$\{aX_0 + bX_1 + cX_2 = 0\}$$

In particular, as lines are paramterised by the triple (a, b, c), we have a correspondence

$$\mathbb{P}^2_{\text{dual}} = \text{lines in } \mathbb{P}^2 \leftrightarrow \mathbb{P}^2$$

The same logic applies for degree d hypersurface in $\mathbb{P}^n$, and we get

$$\text{degree } d \text{ hypersufaces in } \mathbb{P}^n \leftrightarrow \mathbb{P}^N$$

where $N = \binom{n+d}{d} - 1$. However, there is something wrong with this picture. Some polynomials are of the form

$$f = f_1^2 f_2$$

But in this case,

$$\mathbb{V}(f) = \mathbb{V}(f_1 f_2)$$

and $f_1 f_2$ is (in general) not of degree d.

A solution Take $U_d \subseteq \mathbb{P}^N$, where

$$[f] \in U_d \iff f \text{ has no repeated factors}$$

But U_d is not compact.

Output of scheme theory for a fixed projective space $\mathbb{P}^n$, we obtain a space

$$\text{subvarieties of } \mathbb{P}^n \leftrightarrow \text{Var}(\mathbb{P}^n) \subsetneq \text{Hilb}(\mathbb{P}^n) \leftrightarrow \text{ subschemes of } \mathbb{P}^n$$

where $\text{Hilb}(\mathbb{P}^n)$ is the Hilbert scheme of $\mathbb{P}^n$, which is compact in the Euclidean topology. That is, the limit of varieties need not be a variety, but limits of schemes are always schemes.

In scheme theory,

$$\mathbb{V}(X_0 + X_1 + X_2) \text{ and } \mathbb{V}((X_0 + X_1 + X_2)^2)$$

are not isomorphic as schemes.

Weil conjectures Fix $f \in \mathbb{Z}[X_0, \ldots, X_{n+1}]$ homogeneous. We have

$$X = \mathbb{V}(f) \subseteq \mathbb{P}^{n+1}_{\mathbb{C}}$$

We will assume that X is smooth. That is, X is a (complex) manifold. In particular, X is a compact topological space, and so we have numbers $b_0(X), \ldots, b_{2n}(X)$ called the Betti numbers, where

$$b_i(X) = \text{rank}(H_i(X, \mathbb{Z}))$$

In particular, we have the Euler characteristic $\chi(X)$.

On the other hand, fix a prime number p, and let

$$N_m = \text{number of solutions to } f \text{ in } \mathbb{F}_{p^m} = \left|X(\mathbb{F}_{p^m})\right|$$

Define the Weil ζ function

$$\zeta(X; t) = \exp\left(\sum_m \frac{N_m}{m} t^m\right)$$

Theorem 0.1 (Weil conjectures, Grothendieck).

1. $\zeta(X; t)$ is a rational function in t
2. Moreover,
$$\zeta(X; t) = \frac{P_0(t)P_2(t)\cdots P_{2n}(t)}{P_1(t)P_3(t)\cdots P_{2n-1}(t)}$$
where $\deg(P_i(t)) = b_i(X)$.

Upshot: The proof is (fundamentally) via scheme theory. We need a space $\mathfrak{X}$ which interpolates geometry over $\mathbb{C}$ and geometry over finite fields.

Lecture 2

1 Beyond varieties

1.1 Summary of classical algebraic geometry

Let $k = \overline{k}$ be an algebraically closed field. Define the *affine space*

$$\mathbb{A}_k^n = k^n$$

as a set.

An *affine variety* is a subset

$$V = \mathbb{V}(S) \subseteq \mathbb{A}_k^n$$

for some $S \subseteq k[x_1, \ldots, x_n]$. Note that $\mathbb{V}(S) = \mathbb{V}(\langle S \rangle)$, where $\langle S \rangle$ is the ideal generated by S. By the Hilbert basis theorem, or equivalently, $k[x_1, \ldots, x_n]$ is Noetherian, $\langle S \rangle$ is generated by finitely many elements. Moreover,

$$\mathbb{V}(I) = \mathbb{V}\left(\sqrt{I}\right)$$

where I is the *radical* of I, given by

$$\sqrt{I} = \{f \mid f^m \in I \text{ for some } m \in \mathbb{N}\}$$

Given varieties $V \subseteq \mathbb{A}_k^n$, $W \subseteq \mathbb{A}_k^m$, a *morphism* $\varphi : V \to W$ is a function $V \to W$, such that if we write $\varphi = (f_1, \ldots, f_m)$, each f_i is a restriction of a polynomial in $k[x_1, \ldots, x_n]$. An *isomorphism* $\varphi : V \to W$ is a morphism, such that $\psi : W \to V$ also a morphism, $\psi \circ \varphi = \mathrm{id}_V$, $\varphi \circ \psi = \mathrm{id}_W$.

The basic correspondence is

$$\frac{\text{affine varieties over } k}{\text{isomorphism}} \leftrightarrow \text{finitely generated } k\text{-algebras } A \text{ without nilpotent elements}$$

How? Given a variety V (representing an isomorphism class of such), then we can write $V = \mathbb{V}(I)$. Moreover, we can assume that I is a radical ideal. We map

$$V \mapsto \frac{k[x_1, \ldots, x_n]}{I}$$

Conversely, if A is a finitely generated nilpotent-free k-algebra, then by definition,

$$A \cong \frac{k[y_1, \ldots, y_n]}{J}$$

where J is a radical ideal.

Note that we need to check that both maps are well-defined.

Notation 1.1. The algebra associated to V is (classically) denoted $k[V]$, and called the *coordinate ring of* V.

Compatibility of morphisms

Note that we have a correspondence

$$\{\text{morphisms } V \to W\} \leftrightarrow \{k\text{-algebra homomorphisms } k[W] \to k[V]\}$$

Zariski topology

Suppose $V = \mathbb{V}(I) \subseteq \mathbb{A}_k^n$ is an affine variety, with coordinate ring $k[V]$. The *Zariski topology* has closed sets

$$V \cap \mathbb{V}(S)$$

where $S \subseteq k[x_1, \ldots, x_n]$.

If $V \cong W$ as varieties, then the topological spaces with the Zariski topology are also homeomorphic.

Nullstellensatz

Fix a variety V, and let $k[V]$ be its coordinate ring. Given a point $p \in V$, we can produce a homomorphism

$$\begin{aligned} \mathrm{ev}_p : k[V] &\to k \\ \mathrm{ev}_p(f) &= f(p) \end{aligned}$$

Moreover, ev_p is surjective, for example, by taking the constant functions. With this,

$$\mathfrak{m}_p = \ker(\mathrm{ev}_p) \trianglelefteq k[V]$$

is a maximal ideal. So we get a map

$$\{\text{points } p \in V\} \to \{\text{maximal ideals in } k[V]\}$$

Hilbert's Nullstellensatz $\implies$ the above map is a bijection.
Therefore, points of V and maximal ideals in $k[V]$ are "the same thing".

1.2 Limitations

What is an abstract variety?

That is, what is a topological space X, such that we have an open cover $\{U_i\}$, where each U_i is an affine variety, which is compatible on overlaps? For example, we have projective space $\mathbb{P}^n$.

Example 1.2 (non-algebraically closed fields)
Take the ideal

$$I = \left\langle x^2 + y^2 + 1 \right\rangle \trianglelefteq \mathbb{R}[x, y]$$

Observe $\mathbb{V}(I) = \varnothing \subseteq \mathbb{R}^2$, but I is prime, so radical. Hence the Nullstellensatz fails in this case.

On what topological space is $R[x, y]/\left\langle x^2 + y^2 + 1 \right\rangle$ "naturally" the set of functions?

Or more generally, $\mathbb{Z}, \mathbb{Z}[x]$ and so on.

Example 1.3 (why restrict to radical ideals, or nilpotent free algebras)
Let $C = \mathbb{V}(y - x^2) \subseteq \mathbb{A}^2_k$, and $D = \mathbb{V}(y) \subseteq \mathbb{A}^2_k$. Then

$$C \cap D = \mathbb{V}(x, y) = \{(0, 0)\}$$

If we set $D_\delta = \mathbb{V}(y - \delta)$, then $C \cap D_\delta$ is two points, for all $\delta \neq 0$.

Lecture 3

1.3 Spectrum of a ring

Let A be a (commutative, unital) ring.

Definition 1.4 (Zariski spectrum)
The *Zariski spectrum of* A is

$$\operatorname{Spec} A = \{\mathfrak{p} \trianglelefteq A \mid \mathfrak{p} \text{ is a prime ideal}\}$$

Given a ring homomorphism $\varphi : A \to B$, we have a induced map (of sets)

$$\begin{aligned} \varphi^{-1} : \operatorname{Spec}(B) &\to \operatorname{Spec} A \\ \mathfrak{q} &\mapsto \varphi^{-1}(\mathfrak{q}) \end{aligned}$$

Warning: This would fail if we considered maximal ideals instead. That is, the preimage of a maximal ideal need not be maximal.

Given $f \in A$ and $\mathfrak{p} \in \operatorname{Spec} A$, we have an induced

$$\overline{f} \in A/\mathfrak{p}$$

by taking the quotient. Informally, we can evaluate any $f \in A$ at points $\mathfrak{p} \in \operatorname{Spec}(A)$, with the caviat that the codomain of the evaluation map depends on $\mathfrak{p}$.

Example 1.5

Let $A = \mathbb{Z}$, then

$$\operatorname{Spec}(\mathbb{Z}) = \{p \mid p \text{ a prime number}\} \cup \{0\}$$

Choose an element, say $132 \in \mathbb{Z}$. Given a prime p, we can look at $132 \mod p \in \mathbb{Z}/p$.

Takeaway:

$$\operatorname{Spec}(\mathbb{Z}) \rightsquigarrow \text{space}$$

$$132 \in \mathbb{Z} \rightsquigarrow \text{a function}$$

$$132 \mod p \rightsquigarrow \text{value of the function at } p$$

Example 1.6

$A = \mathbb{R}[x]$, then

$$\operatorname{Spec} \mathbb{R}[x] = \frac{\mathbb{C}}{\text{complex conjugation}} \cup \{0\}$$

Exercise: Draw $\operatorname{Spec} \mathbb{Z}[x]$ and $\operatorname{Spec} k[x]$ for any field k.

Example 1.7 (for sanity)

If $A = \mathbb{C}[x]$, then

$$\operatorname{Spec} A = \mathbb{C} \cup \{0\}$$

Given $a \in \mathbb{C}$, we have the maximal ideal $\langle z - a \rangle$.

1.4 Topology

Fix $f \in A$, then define

$$\mathbb{V}(f) = \{\mathfrak{p} \in \operatorname{Spec}(A) \mid f \equiv 0 \pmod{\mathfrak{p}}\} \subseteq \operatorname{Spec}(A)$$

or equivalently, $f + \mathfrak{p} = 0 \in A/\mathfrak{p}$, or $f \in \mathfrak{p}$. Similarly, for $J \trianglelefteq A$ an ideal,

$$\mathbb{V}(J) = \{\mathfrak{p} \in \operatorname{Spec}(A) \mid J \subseteq \mathfrak{p}\}$$

Proposition 1.8. The sets $\mathbb{V}(J) \subseteq \operatorname{Spec} A$ ranging over all ideals $J \trianglelefteq A$, form the closed sets of a topology on $\operatorname{Spec} A$. This topology is called the *Zariski topology*.

Proof. Easy facts that $\varnothing = \mathbb{V}(1)$ and $\operatorname{Spec} A = \mathbb{V}(0)$ are closed. Since

$$\mathbb{V}\left(\sum_\alpha I_\alpha\right) = \bigcap_\alpha \mathbb{V}(I_\alpha)$$

the intersections of closed sets is closed. Finally,

$$\mathbb{V}(I_1 \cap I_2) = \mathbb{V}(I_1) \cup \mathbb{V}(I_2)$$

$\supseteq$ is clear, conversely, $I_1 I_2 \subseteq I_1 \cap I_2 \subseteq \mathfrak{p}$, then by primality of $\mathfrak{p}$, $I_1 \subseteq \mathfrak{p}$ or $I_2 \subseteq \mathfrak{p}$. □

Example 1.9

Consider $\operatorname{Spec}\mathbb{C}[x,y]$. A few observations:

- $0 \in \operatorname{Spec}\mathbb{C}[x,y]$ is dense in the Zariski topology. That is,
$$\overline{\{\langle 0\rangle\}} = \operatorname{Spec}\mathbb{C}[x,y]$$
since every prime ideal contains 0.
- Consider the prime ideal $\langle y^2 - x^3\rangle$. Consider a maximal ideal
$$\mathfrak{m}_{a,b} = \langle x-a, y-b\rangle$$
When is $\mathfrak{m}_{a,b} \in \overline{\{\langle y^2-x^3\rangle\}}$? This holds if and only if $b^2 = a^3$. See examples sheet 1.

Remark 1.10. Points are not closed.

1.5 Functions on opens

Let $f \in A$. Define the *distinguished open corresponding to* f to be
$$U_f = \operatorname{Spec}(A) \setminus \mathbb{V}(f)$$

Example 1.11

If $A = \mathbb{C}[x]$, then
$$\operatorname{Spec} A = \mathbb{C} \cup \{\langle 0\rangle\}$$
Take $f = x$. Recall that we have a bijection
$$\begin{aligned} \operatorname{Spec}(A) &\leftrightarrow \mathbb{C} \cup \{\langle 0\rangle\} \\ (x-a) &\mapsfrom a \in \mathbb{C} \\ 0 &\mapsfrom 0 \end{aligned}$$
Then $\mathbb{V}(x) = \{\mathfrak{p} \in \operatorname{Spec} A \mid x \in \mathfrak{p}\} = \{\langle x\rangle\}$. So
$$U_x = \operatorname{Spec}(A) \setminus \{\langle 0\rangle\}$$
As a picture, $\operatorname{Spec}(\mathbb{C}[x])$ is a line (corresponding to $\mathbb{C}$) with a "generic point" $\{0\}$.

More generally, suppose we fix $a_1, \dots, a_r \in \mathbb{C}$, with
$$U = \operatorname{Spec}(A) \setminus \{\langle x - a_i\rangle\}_{i=1}^r$$
then $U = U_f$ where $f = \prod_{i=1}^r (x-a_i)$.

Lemma 1.12. The distinguished opens U_f for all $f \in A$ form a basis for the Zariski topology on $\operatorname{Spec} A$.

Proof. Examples sheet 1. □

A bit of commutative algebra: Given $f \in A$, the *localisation of* A *at* f is
$$A_f = \frac{A[x]}{\langle xf - 1\rangle} = A\left[\frac{1}{f}\right]$$

Lemma 1.13. The distinguished open $U_f \subseteq \operatorname{Spec}(A)$ is naturally homeomorphic to $\operatorname{Spec}(A_f)$, via the ring homomorphism

$$j : A \to A_f$$

which induces

$$j^{-1} : \operatorname{Spec}(A_f) \to \operatorname{Spec} A$$

Proof. Primes in A_f are in bijection with primes of A, which miss f, via j^{-1}. To see this, the preimage of a prime ideal is prime, and so if $\mathfrak{q} \trianglelefteq A_f$ is prime, then $j^{-1}(\mathfrak{a}) \trianglelefteq A$ is prime. For the converse, given $\mathfrak{p} \trianglelefteq A$, define

$$\mathfrak{p}_f = j(\mathfrak{p})A_f = \langle j(\mathfrak{p}) \rangle$$

Claim 1.14. $\mathfrak{p}_f$ is a prime exactly when $f \notin \mathfrak{p}$.

Proof of claim. To see this, f is a unit in A_f, and so if $f \in \mathfrak{p}$, then $\mathfrak{p}_f$ contains a unit, and so $\mathfrak{p}_f = 1$.

But if $f \notin \mathfrak{p}$, then

$$\frac{A_f}{\mathfrak{p}_f} \cong \left(\frac{A}{\mathfrak{p}}\right)_{\overline{f}}$$

where $\overline{f} = f + \mathfrak{p}$. So

$$(A/\mathfrak{p})_{\overline{f}} \subseteq \operatorname{Frac}(A/\mathfrak{p})$$

and so it is an integral domain. □

Finally, we can check that this defines a bijection, but this is just basic properties of ideals. □

Facts about distinguised opens

- $U_f \cap U_g = U_{fg}$.
- $U_{f^n} = U_f$ for all $n \geq 1$.
- The rings A_f and A_{f^n} (for $n \geq 1$) are isomorphic. To see this,

  $$A_f = \frac{A[x]}{\langle xf - 1 \rangle} \quad \text{and} \quad A_{f^n} = \frac{A[y]}{\langle yf^n - 1 \rangle}$$

  The isomorphism is given by

  $$y \mapsto x^n \quad \text{and} \quad x \mapsto f^{n-1}y$$

  which define inverse maps.
- Containment: $U_f \subseteq U_g$ if and only if f^n is a multiple of g for some $n \geq 1$. (Recall if $f = gf'$, then certainly $U_f \subseteq U_g$)

  Proof. The "if" direction is clear. Conversely, if $U_f \subseteq U_g$. That is,

  $$\mathbb{V}(f) \supseteq \mathbb{V}(g)$$

  The set $\mathbb{V}(f)$, by definition, is the set of all primes, containing f.

  Claim 1.15. $\sqrt{\langle f \rangle} \subseteq \sqrt{\langle g \rangle}$.

  To see this, the radical of an ideal I is the intersection of all primes containing I, which we will see in Commutative Algebra. □

Foreshadowing

Fix A, we've made an assignment

$$\begin{aligned} \text{distinguised opens in } \operatorname{Spec} A &\to \text{rings} \\ U_f &\mapsto A_f \end{aligned}$$

The association is functorial, that is, if $U_{f_1} \subseteq U_{f_2}$, then we can assume $f_1^n = f_2 f_3$, so $U_{f_1} = U_{f_1^n} = U_{f_2 f_3} \subseteq U_{f_2}$, and so there is a homomorphism

$$A_{f_2} \to A_{f_1}$$

which is the restriction map.

Can we extend this association to all open sets? See notes.

2 Sheaves

2.1 Presheaves

Let X be a topological space.

Definition 2.1 (presheaf)

A *presheaf!of abelian groups* is an association

$$\begin{aligned} \{\text{Open set in } X\} &\leftrightarrow \text{Abelian Groups} \\ U &\mapsto \mathcal{F}(U) \end{aligned}$$

and for $U \subseteq V$ opens, we have a homomorphism

$$\operatorname{res}_U^V : \mathcal{F}(V) \to \mathcal{F}(U)$$

such that

1. $\operatorname{res}_U^U = \mathrm{id}$,
2. $\operatorname{res}_U^V \operatorname{res}_V^W = \operatorname{res}_U^W$ for $U \subseteq V \subseteq W$.

Example 2.2

For any space X, take

$$\mathcal{F}(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$$

with $\operatorname{res}_U^V(f) = f|_U$.

We can make an analogous definition for sheaves of rings, sets, R-modules, and so on.

Definition 2.3 (morphism of presheaves)

A *morphism* $\varphi : \mathcal{F} \to \mathcal{G}$ *of presheaves* on X is, for each $U \subseteq X$ open, a homomorphism

$$\varphi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$$

compatible with restrictions, that is, the diagram

$$\begin{array}{ccc} \mathcal{F}(V) & \xrightarrow{\varphi(V)} & \mathcal{G}(V) \\ \downarrow{\scriptstyle \operatorname{res}_U^V} & & \downarrow{\scriptstyle \operatorname{res}_U^V} \\ \mathcal{F}(U) & \xrightarrow[\varphi(U)]{} & \mathcal{G}(U) \end{array}$$

> commutes, for all $U \subseteq V$.

In terms of category theory, the opens in X form a category, with morphisms given by inclusions. In this case, a presheaf is a contravariant functor $\mathrm{Opens}(X) \to \mathrm{Ab}$, and a morphism is a natural transformation.

A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ is *injective* (resp. *surjective*) if $\varphi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ is injective (resp. surjective) for all $U \subseteq X$.

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